k-Plan Planning Report

Plan Summary

Subject Information	
Name:	**CITRUS-offline SUB-04**
ID1 / ID2:	**TN / SV**
Date of Birth:	**12 Mar 2026**
Age:	**0**
Gender:	**Not Known (not recorded)**
Comments:	

Plan Information	
Plan Label:	**sub-04_Tx-2_R-pos-4_L-pos-5_offline_I-6_EXP_FINAL**
Plan Comments:	**RL sgACC focused 5Hz thermal**
Transducer Model:	**CITRUS-V1-FULL-REV-7**
Transducer ID:	**CITRUS-V1-FULL-REV-7**
Transducer Manufacturer:	**Fraunhofer IBMT**
Transducer Type:	**Multi-Element Array**
Number of Sonications:	**2**

Label	Frequency	$P_{tp, target}$	$I_{pa, target}$*	$P_{sptp, in situ}$	$I_{sppa, in situ}$*	MI_{tc}
Tx-2_R_pos-4_I-6_exp	**286 kHz**	**424 kPa**	**6 W/cm^2**	**914 kPa**	**27.9 W/cm^2**	**1.71**
Tx-2_L_pos-5_I-6_exp	**286 kHz**	**424 kPa**	**6 W/cm^2**	**860 kPa**	**24.7 W/cm^2**	**1.48**

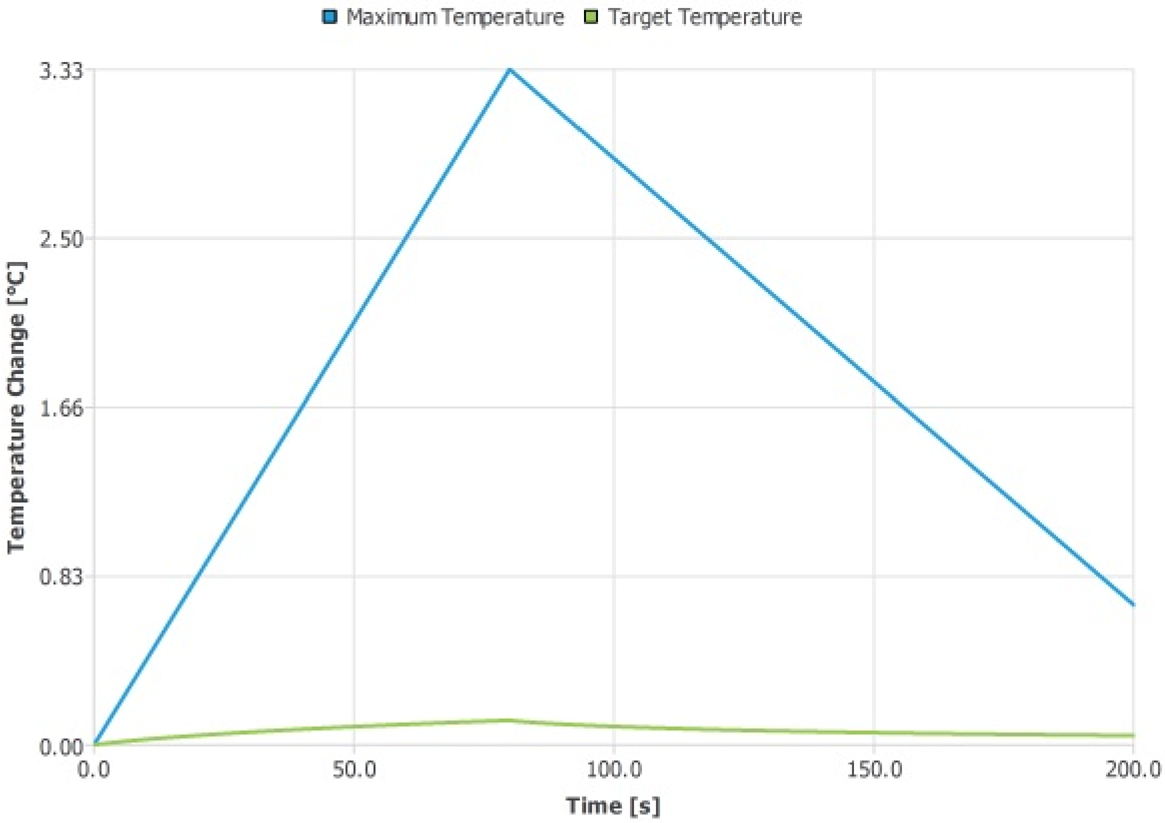
Label	Target ΔT	Max ΔT	Target TD	Max TD
Tx-2_R_pos-4_I-6_exp	**0.122 °C**	**3.33 °C**	**<0.1 CEM**	**<0.1 CEM**
Tx-2_L_pos-5_I-6_exp	**0.15 °C**	**4.51 °C**	**<0.1 CEM**	**<0.1 CEM**

Sonication 1: Tx-2_R_pos-4_I-6_exp

System Parameters	
Driving Frequency:	**286 kHz**
Target Position:	**128 mm, 171 mm, 159 mm**
Cooling Time:	**120 s**

	Duration	Repetition Interval	Ramp Shape	Ramp Duration	$I_{ta, target}$*	$I_{spta, in situ}$*
Pulse	**25 ms**	**200 ms**	**Tukey (Pressure/Voltage)**	**5 ms**	**563 mW/cm^2**	**2.61 W/cm^2**
Pulse Train	**80 s**	**80 s**	**No Ramp**	**0 s**	**563 mW/cm^2**	**2.61 W/cm^2**

Regional Peak Values			
	P_{sptp}	Max ΔT	Max Thermal Dose
background	**739 kPa**	**229 m°C**	**<0.1 CEM**
head	**914 kPa**	**2.67 °C**	**<0.1 CEM**
skull	**866 kPa**	**3.33 °C**	**<0.1 CEM**

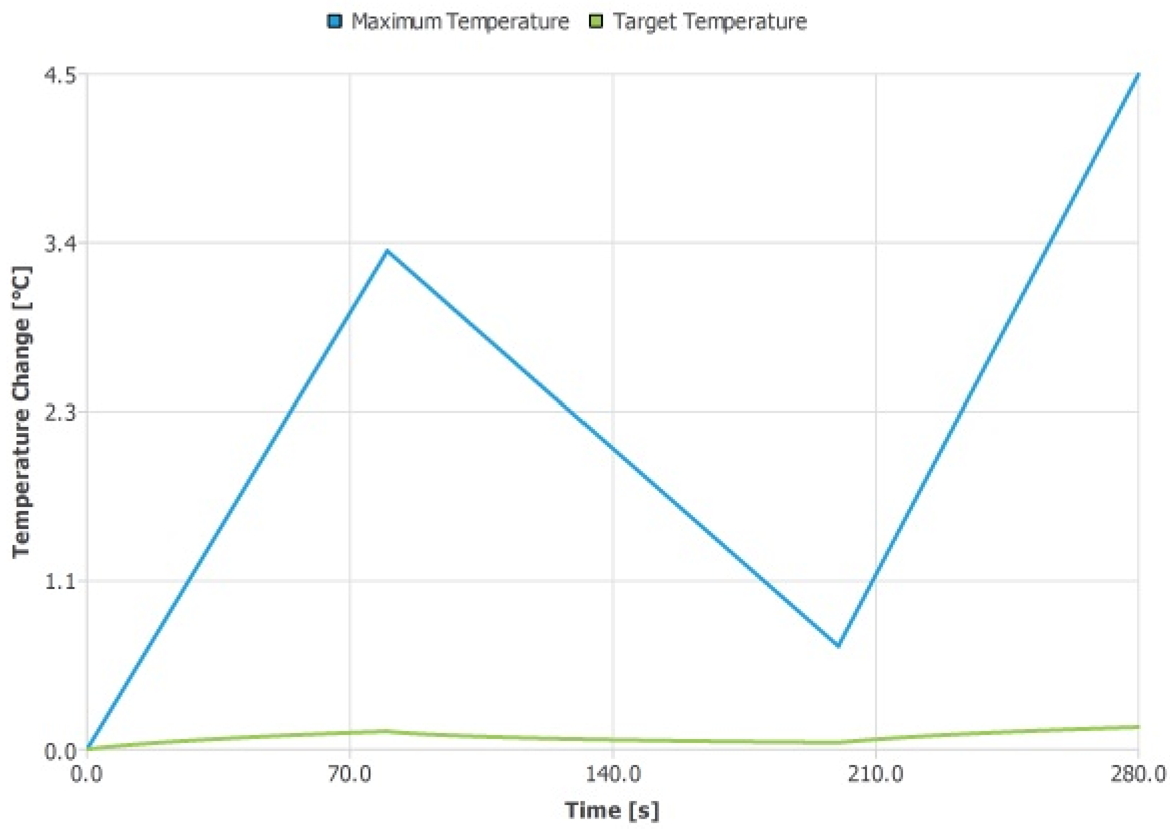


Sonication 2: Tx-2_L_pos-5_I-6_exp

System Parameters	
Driving Frequency:	**286 kHz**
Target Position:	**122 mm, 172 mm, 159 mm**
Cooling Time:	**0 s**

	Duration	Repetition Interval	Ramp Shape	Ramp Duration	$I_{ta, target}$*	$I_{spta, in situ}$*
Pulse	**25 ms**	**200 ms**	**Tukey (Pressure/Voltage)**	**5 ms**	**563 mW/cm²**	**2.31 W/cm²**
Pulse Train	**80 s**	**80 s**	**No Ramp**	**0 s**	**563 mW/cm²**	**2.31 W/cm²**

Regional Peak Values			
	P_{sptp}	Max ΔT	Max Thermal Dose
background	**714 kPa**	**251 m°C**	**<0.1 CEM**
head	**789 kPa**	**3.16 °C**	**<0.1 CEM**
skull	**860 kPa**	**4.51 °C**	**<0.1 CEM**



k-Plan Settings

k-Plan Information	
Plan Created Using:	**k-Plan TPM v1.2.0**
Plan Executed Using:	**k-Plan SEM v1.2.1**
Plan Executed On:	**18 March 2026**
Report Generated Using:	**k-Plan TPM v1.2.0**
Report Generated On:	**2 April 2026**
Report Generated By:	**Suhas Vijayakumar**

Simulation Settings	
Domain Size:	**215 mm × 202 mm × 165 mm**
Grid Spacing:	**897 um**
Grid Points Per Wavelength:	**6**
Output Downsampling Factor:	**1**
Grid Traversals:	**2**
Run Thermal Simulations:	**Yes**
Linked Sonications:	**Yes**
Material Property Conversion Method:	**Head CT**

*The temporal-average (ta) intensity values account for any ramping, while the pulse-average (pa) intensity values do not account for ramping.